

PRANAV J. KOLI

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EDUCATION

University of Maryland, College Park

A. James Clark School of Engineering
ME in Robotics Engineering

Aug 2025 – May 2027

GPA 3.9/4

University of Mumbai

Agnel Charities' Fr. C. Rodrigues Institute of Technology
BE in Mechanical Engineering

Aug 2020 – May 2024

GPA 9.59/10

TECHNICAL SKILLS

- **Modeling:** AutoCAD, Solidworks (CSWA), Inventor, Fusion 360
- **Robotics & Control:** ROS1/ROS2, MoveIt, MPC, LQR, SLAM
- **Programming:** MATLAB, C/C++, Python, OpenCV, PyTorch, CUDA, G&M
- **Simulation:** ANSYS, Gazebo, Issac Sim, MuJoCo, PyBullet, Rviz, Webots, CARLA, Duality AI (Falcon)
- **Others:** Docker, Git, Linux, Blender, Eagle, Slicer 3D

RELEVANT WORK EXPERIENCE

Inception Robotics LLC

Robotics Intern

College Park, MD

Jun 2026 – Aug 2026

- Evaluate hardware and mechatronic design options for the MedSort robotic device, including controllers, actuators, motors, sensors, and mechanical components for industrial certification.
- Evaluate electronic components and motor/servo drives using industrial communication protocols such as EtherCAT and CANopen.
- Evaluate different trajectory generation algorithms for robotic motion planning, train vision models on CUDA, simulate the process on Issac Sim.

Apex Precision Mechatronics Pvt. Ltd.

Design Engineer

Mumbai, India

Aug 2024 – Jul 2025

- Designing of linear motion control and industrial automation systems for user specified application. (linear rails, actuators, lead & ball screws, belt drives)
- Selection assistance as per the customer's requirement for product development and optimizing current designs.
- Creating and validating technical documentations as per industrial standards.

Mazagon Docks Shipbuilders Ltd.

Intern

Mumbai, India

Jun 2022 – Jul 2022

- Learned various manufacturing processes involved for production of a naval ship along with power and piping systems within it.
- Designing of ships by outfitting several compartments and understanding complex drawings of general arrangement.

Indian Oil Corporation Ltd.

Intern

Mumbai, India

Dec 2022 – Jan 2023

- Learned different production processes for creating petroleum products along with operation of machineries used in plants.
- Studied wide range of petroleum products as per their characteristics and applications.

ACADEMIC PROJECT EXPERIENCE

Frenet Frame Polynomial Lattice Planner for Highway Autonomous Driving Mar 2026 – May 2026

- Engineered an end-to-end autonomous driving motion planning stack in the CARLA simulator, covering reference path generation, Frenet coordinate transformations, candidate trajectory generation, safety filtering, cost optimization, and vehicle control.
- Filtered unsafe paths using acceleration and collision bounds, selecting optimal trajectories via a multi-objective cost function penalizing jerk, speed error, and lane offset.
- Integrated Pure Pursuit steering and PID speed controllers to translate optimal trajectories into throttle and steering commands.

Pick-and-Place Manipulation Stack for Franka Panda Robot Mar 2026 – Apr 2026

- Developed an end-to-end Pick and Place pipeline for the Franka Emika Panda arm using ROS/MoveIt for trajectory planning and execution.
- Configured end-effector controllers and grasp poses to execute precise object pick, move, and place sequences in simulation/hardware.
- Integrated 3D scene perception and planning scene monitors to dynamically detect obstacles and prevent arm collisions.

Adaptable Real-Time Surface Painting Robot Nov 2025 – Dec 2025

- Architected an autonomous painting system using a 6-DoF UR5 articulated arm, to reconstruct unknown surface geometries and autonomously generate adaptive zig-zag trajectories in real-time.
- Developed a perception-driven control framework in ROS2 and C++, implementing a numerical Jacobian pseudo-inverse solver for inverse kinematics and Principal Component Analysis (PCA) to extract surface normals for high-precision tool orientation.

Implementation of Lateral Control Algorithms for Autonomous Vehicles Sep 2025 – Oct 2025

- Evaluated and implemented three lateral control algorithms—Model Predictive Control (MPC), Pure Pursuit (PP), and Linear Quadratic Regulator (LQR)—using the CARLA simulator and Python.
- Optimized tracking precision and real-time efficiency by developing a latency-compensated LQR controller and a linearized MPC strategy.

Design & Development of an Inboard Braking System for a RWD ATV Jun 2023 – May 2024

- Designed and fabricated unit of Inboard Braking by means of mounting braking system on the output shaft of gearbox improving weight dynamics of the rear side of the vehicle.
- Successfully tested and validated the enhanced stability and performance of vehicle.

INVOLVEMENT

Agnels Robotics Club Jul 2021 – May 2024

- Agnels Robotics Team member in the mechanical division.
- Participated in Doordarshan Robocon India National Competition in 2022 & 2023 hosted by IIT Delhi.
- Achieved All India Rank 15 (AIR 15) in Robocon 2022 among the 42 qualified teams for stage 3.
- Planned and conducted workshops and seminars related to robotics.

DECLARATION

I hereby declare that all of the above information is true and to the best of my knowledge and I bear the responsibility for the correctness of the above-mentioned.

Date:

Signature